

Cover Page

Lab 4	<u>Footprinting & Reconnaissance</u>
Lab 5	<u>Host Discovery</u>
Duration	4 hours

Module	Professional Penetration Testing
Instructor	Phillip Fitzpatrick

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Date	23/02/2026	Submission Date	02/03/2026
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University Declaration & Use of AI

University Declaration

I confirm that this submission is my own work, except where clearly indicated. I have acknowledged all sources used. I understand the university's policies on academic integrity and plagiarism.

Use of AI Tools Declaration

I had not used AI at any stage of this assessment.

Signed: Danyil Tymchuk

Date: 23/02/2026

Lab 4: Footprinting & Reconnaissance

Understanding Search engines and Google dorking

Introduction to Shodan.io

Introduction to Censys.io

whois and NsLookup

TheHarvester

Recon-ng

FOCA

Maltego

OSINT Framework

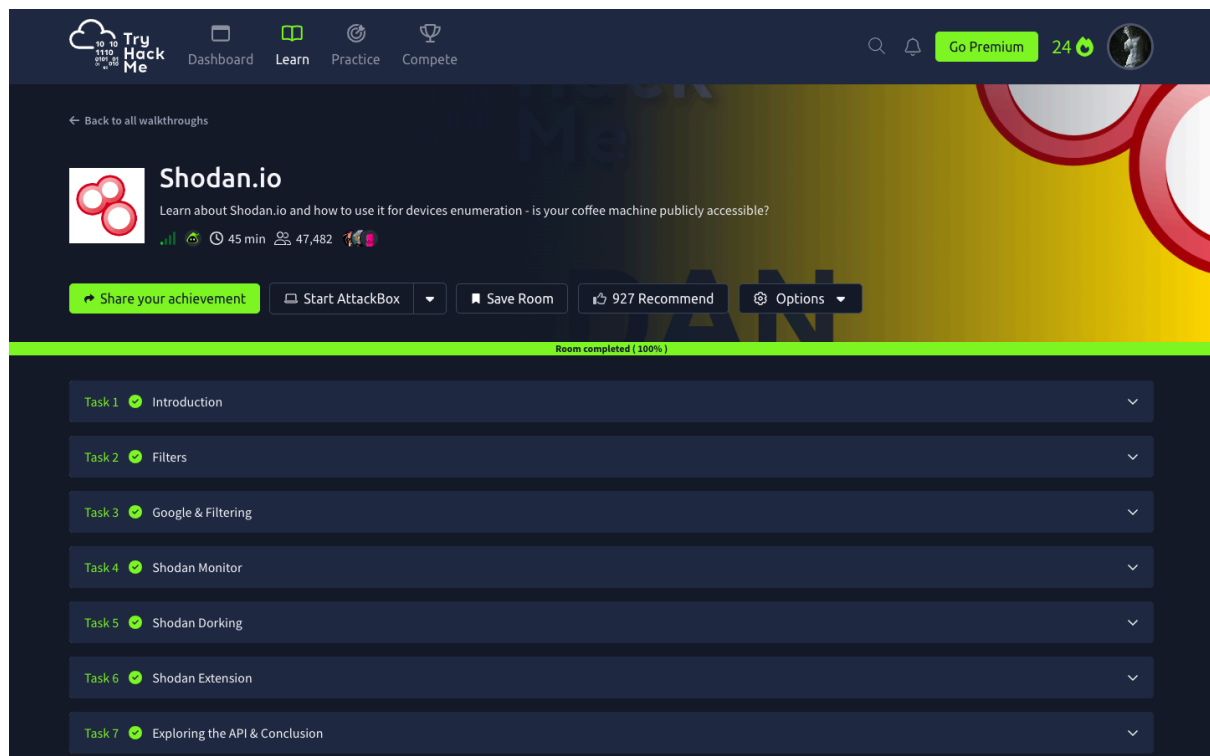
Additional THM Related Rooms

Understanding Search engines and Google dorking

The screenshot displays the TryHackMe interface for the 'Google Dorking' room. At the top, the navigation bar includes 'TryHackMe', 'Dashboard', 'Learn', 'Practice', and 'Compete'. The room title 'Google Dorking' is prominently displayed, along with a subtitle and a description. Below the title, there are buttons for 'Share your achievement', 'Start AttackBox', 'Save Room', and 'Recommend'. A green bar indicates 'Room completed (100%)'. A list of six tasks follows, each marked as completed with a green checkmark. The tasks are: Task 1: Ye Ol' Search Engine, Task 2: Let's Learn About Crawlers, Task 3: Enter: Search Engine Optimisation, Task 4: Beepboop - Robots.txt, Task 5: Sitemaps, and Task 6: What is Google Dorking?.

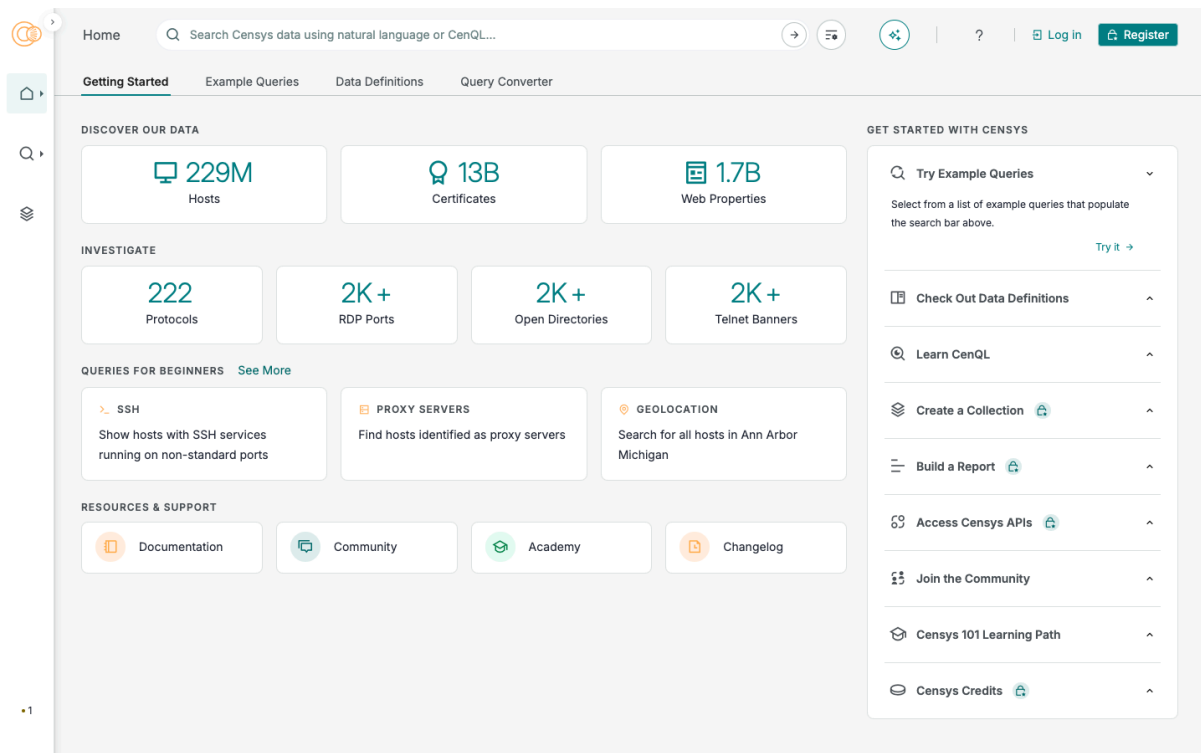
☒ Complete the Google Dorking Room on THM

Introduction to Shodan.io



- ☒ Complete the Shodan.io Room on THM

Introduction to Censys.io



- ☒ Explore the Censys.io website

whois and Nslookup

whois example.com

```
Domain Name: EXAMPLE.COM
Registry Domain ID: 2336799_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.iana.org
Registrar URL: http://res-dom.iana.org
Updated Date: 2026-01-16T18:26:50Z
Creation Date: 1995-08-14T04:00:00Z
Registry Expiry Date: 2026-08-13T04:00:00Z
Registrar: RESERVED-Internet Assigned Numbers Authority
Registrar IANA ID: 376
Registrar Abuse Contact Email:
Registrar Abuse Contact Phone:
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: ELLIOTT.NS.CLOUDFLARE.COM
Name Server: HERA.NS.CLOUDFLARE.COM
DNSSEC: signedDelegation
DNSSEC DS Data: 2371 13 2 C988EC423E3880EB8D08A46FE06CA230EE23F35B578D64E78B29C3E1C83D245A
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-03-02T11:12:54Z <<<
```

nslookup example.com 8.8.8.8

```
(kali# kali)-[~]
$ nslookup example.com 8.8.8.8 | less
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
Name:   example.com
Address: 104.18.26.120
Name:   example.com
Address: 104.18.27.120
Name:   example.com
Address: 2606:4700::6812:1a78
Name:   example.com
Address: 2606:4700::6812:1b78

(END)_
```

dig example.com

```
(kali# kali)-[~]
$ dig example.com | less
;; communications error to 8.8.8.8#53: timed out

; <<> Dig 9.20.18-1-Debian <<> example.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<<- opcode: QUERY, status: NOERROR, id: 21191
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags: udp: 512
;; QUESTION SECTION:
;example.com.                IN      A

;; ANSWER SECTION:
example.com.                208     IN      A      104.18.27.120
example.com.                208     IN      A      104.18.26.120

;; Query time: 40 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Mon Mar 02 06:19:56 EST 2026
;; MSG SIZE rcvd: 72

(END)_
```

☒ Explore whois, nslookup, dig

TheHarvester

```

kali@kali:~$ ./theHarvester -d example.com -b duckduckgo | less
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*
* theHarvester
*
*
* theHarvester 4.9.2
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****

[*] Target: example.com

[*] Searching Duckduckgo.

[*] No IPs found.

[*] No emails found.

[*] No people found.

[*] Hosts found: 0

-----
(END)_

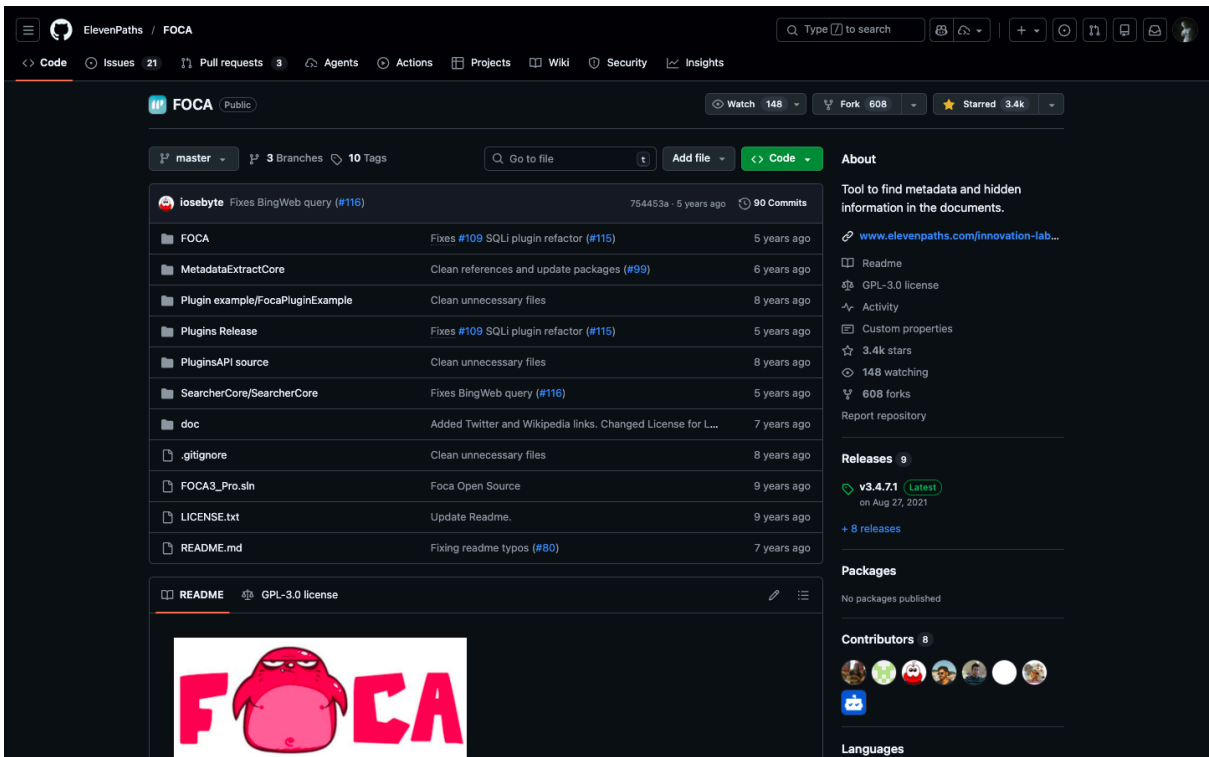
```

☒ Explore theHarvester

Recon-ng

[illegible]☒ Explore Recon-ng

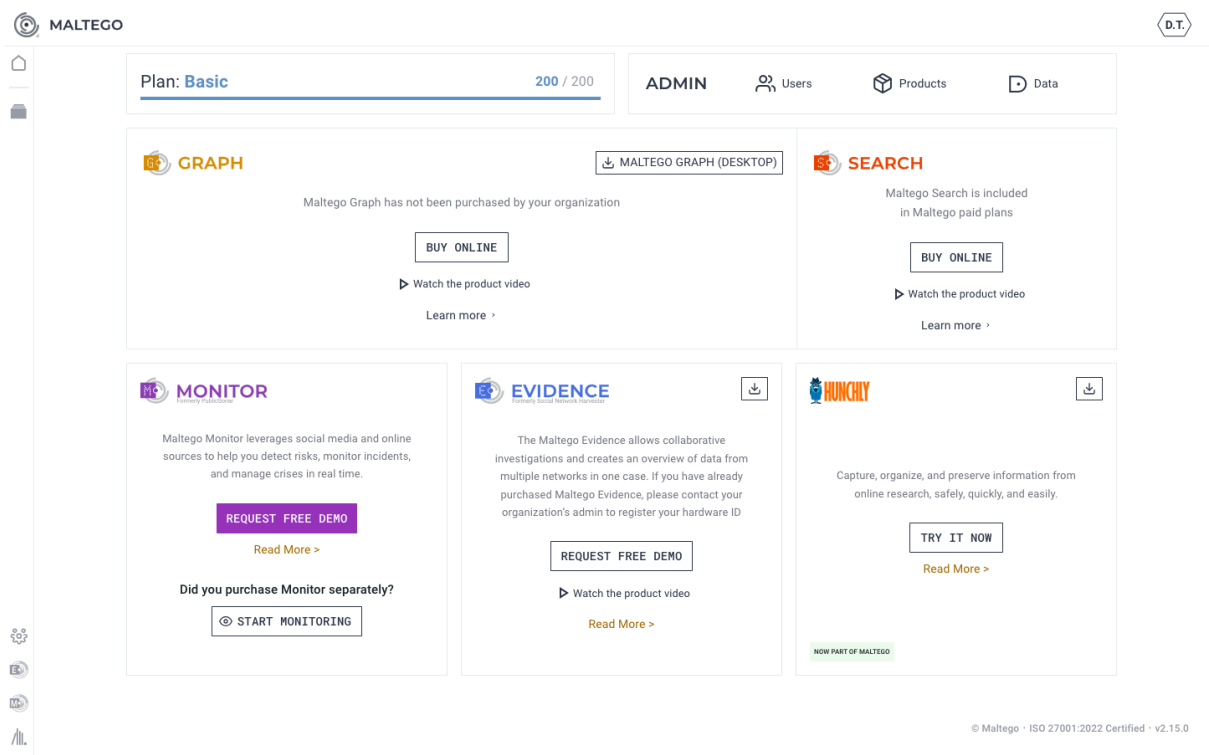
FOCA



The screenshot shows the GitHub repository for FOCA, a tool for finding metadata and hidden information. The repository is owned by ElevenPaths and is public. It has 148 watchers, 608 forks, and 3.4k stars. The repository is currently on the master branch, with 3 branches and 10 tags. The repository contains a README, a LICENSE.txt, and a README.md. The README features a cartoon character and the text "FOCA". The repository is licensed under the GPL-3.0 license. The repository is also available as a package on GitHub Packages.

☒ Explore FOCA

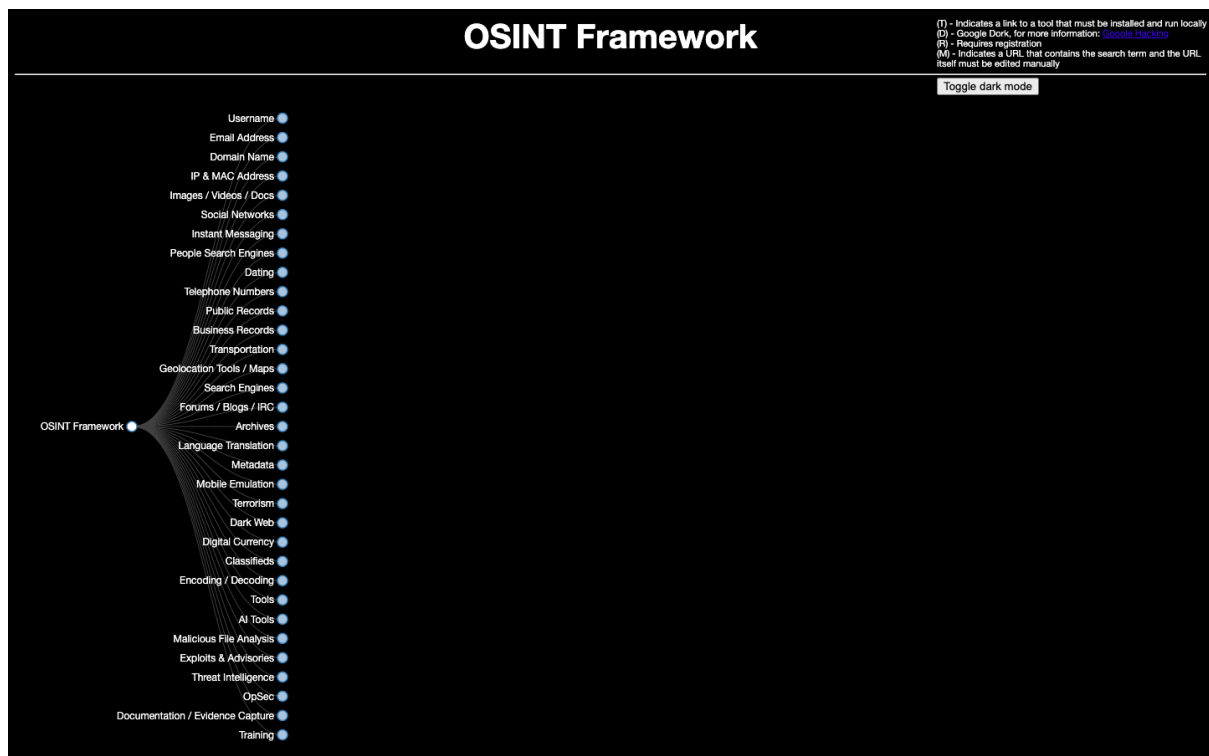
Maltego



The screenshot shows the Maltego website, which is a platform for digital investigation. The website features a navigation bar with links to ADMIN, Users, Products, and Data. The main content area is divided into four sections: GRAPH, SEARCH, MONITOR, and EVIDENCE. Each section has a brief description of the tool and a button to purchase or request a demo. The GRAPH section has a button to download the desktop version. The SEARCH section has a button to buy online. The MONITOR section has a button to request a free demo. The EVIDENCE section has a button to request a free demo. The HUNCHY section has a button to try it now. The website also features a footer with the text "© Maltego · ISO 27001:2022 Certified · v2.15.0".

☒ Explore Maltego

OSINT Framework



☒ Explore OSINT Framework

Additional THM Related Rooms

TryHackMe Dashboard Learn Practice Compete

Cyber Security 101 > Start Your Cyber Security Journey > Search Skills

Search Skills

Learn to efficiently search the Internet and use specialized search engines and technical docs.

60 min 469,613

Share your achievement Start AttackBox Save Room 3855 Recommend Options

Room completed (100%)

- Task 1 Introduction
- Task 2 Evaluation of Search Results
- Task 3 Search Engines
- Task 4 Specialized Search Engines
- Task 5 Vulnerabilities and Exploits
- Task 6 Technical Documentation
- Task 7 Social Media

☒ Complete the Search Skills Room on THM

The screenshot shows the 'Passive Reconnaissance' room completion page on TryHackMe. The header includes the TryHackMe logo, navigation links (Dashboard, Learn, Practice, Compete), a search icon, a 'Go Premium' button, a '24' badge, and a user profile icon. Below the header, there's a 'Back to all walkthroughs' link. The main section features a magnifying glass icon, the title 'Passive Reconnaissance', a description 'Learn about the essential tools for passive reconnaissance, such as whois, nslookup, and dig.', a progress bar, a '60 min' timer, a '236,297' user count, and a '5049 Recommend' button. A green bar indicates 'Room completed (100%)'. Below this, a list of tasks is shown, all marked as completed with green checkmarks:

- Task 1 Introduction
- Task 2 Passive Versus Active Recon
- Task 3 Whois
- Task 4 nslookup and dig
- Task 5 DNSDumpster
- Task 6 Shodan.io
- Task 7 Summary

☒ Complete the Passive Reconnaissance Room on THM

The screenshot shows the 'Active Reconnaissance' room completion page on TryHackMe. The header is identical to the previous screenshot. Below the header, there's a 'Back to all walkthroughs' link. The main section features a magnifying glass icon, the title 'Active Reconnaissance', a description 'Learn how to use simple tools such as traceroute, ping, telnet, and a web browser to gather information.', a progress bar, a '60 min' timer, a '184,984' user count, and a '3567 Recommend' button. A green bar indicates 'Room completed (100%)'. Below this, a list of tasks is shown, all marked as completed with green checkmarks:

- Task 1 Introduction
- Task 2 Web Browser
- Task 3 Ping
- Task 4 Traceroute
- Task 5 Telnet
- Task 6 Netcat
- Task 7 Putting It All Together

☒ Complete the Active Reconnaissance Room on THM

Lab 5: Host Discovery

Understanding the PING command

Understanding the Fping command

When to use ARP scan

Learning to use Nmap

Nmap Live Host Discovery

Learning to use RustScan

Understanding the PING command

1. Type ifconfig to check your IP address. What is your IP address on the virtual network?

192.168.56.2

```
(kali~ kali)~  
[~$ ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.56.2 netmask 255.255.255.0 broadcast 192.168.56.255  
    inet6 fe80::fe01:106d:e45b:f63a prefixlen 64 scopeid 0x20<link>  
    ether 08:00:27:63:b0:05 txqueuelen 1000 (Ethernet)  
    RX packets 2391 bytes 196840 (192.2 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 5873 bytes 361335 (352.8 KiB)  
    TX errors 0 dropped 1 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 841 bytes 37336 (36.4 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 841 bytes 37336 (36.4 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
(kali~ kali)~  
[~$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:63:b0:05 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.56.2/24 brd 192.168.56.255 scope global dynamic noprefixroute eth0  
        valid_lft 83788sec preferred_lft 83788sec  
    inet6 fe80::fe01:106d:e45b:f63a/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever
```

2. What do you think would happen if you ping your network broadcast address?
(Different on windows and Linux) Try it. Why did you get the results you got?

sends ICMP to host/router network (192.168.56.1)

```
(kali~ kali)~  
[~$ ping -b 192.168.56.255  
WARNING: pinging broadcast address  
PING 192.168.56.255 (192.168.56.255) 56(84) bytes of data.  
64 bytes from 192.168.56.1: icmp_seq=1 ttl=64 time=4.24 ms  
64 bytes from 192.168.56.1: icmp_seq=2 ttl=64 time=0.483 ms  
64 bytes from 192.168.56.1: icmp_seq=3 ttl=64 time=0.995 ms  
64 bytes from 192.168.56.1: icmp_seq=4 ttl=64 time=1.20 ms  
64 bytes from 192.168.56.1: icmp_seq=5 ttl=64 time=1.09 ms  
-
```

3. Look at the help for ping. What command would you use to send exactly 100 ping packets to your target machine?

ping -c 100 <target_ip>

```
Options:
<destination>    DNS name or IP address
-s              RTT precision (do not round up the result time)
-a              use audible ping
-A              use adaptive ping
-B              sticky source address
-c <count>       stop after <count> replies
-C              call connect() syscall on socket creation
-D              print timestamps
-d              use SO_DEBUG socket option
-e <identifier>  define identifier for ping session, default is random for
                  SOCK_RAW and kernel defined for SOCK_DGRAM
                  imply using SOCK_RAW (for IPv4 only for identifier 0)
-f              flood ping
-h              print help and exit
-H              force reverse DNS name resolution (useful for numeric
                  destinations or for -f), override -n
-I <interface>  either interface name or address
-i <interval>   seconds between sending each packet
-L              suppress loopback of multicast packets
-l <preload>    send <preload> number of packages while waiting replies
-m <mark>       tag the packets going out
-M <pmtd opt>  define path MTU discovery, can be one of <do|dont|want|probe>
-n              no reverse DNS name resolution, override -H
-O              report outstanding replies
-p <pattern>    contents of padding byte
-q              quiet output
-Q <tc|class>   use quality of service <tc|class> bits
-s <size>       use <size> as number of data bytes to be sent
-S <size>       use <size> as SO_SNDBUF socket option value
-t <ttl>        define time to live
-U              print user-to-user latency
-v              verbose output
-V              print version and exit
-w <deadline>   reply wait <deadline> in seconds
-W <timeout>    time to wait for response

IPv4 options:
-4              use IPv4
-b              allow pinging broadcast
-R              record route
-T <timestamp>  define timestamp, can be one of <tsonly|tsandaddr|tsprespec>

IPv6 options:
-6              use IPv6
-F <flowlabel>  define flow label, default is random
-N <nodeinfo opt> use IPv6 node info query, try <help> as argument

For more details see ping(8).
(END)
```

Understanding the Fping command

- Again look at the help for the fping command. What command would you use to send exactly 100 ping packets to your target machine?

```
fping -c 100 <target_ip>
```

```
(kali㉿kali)-[~]
└─$ fping --help | less
Usage: fping [options] [targets...]

Probing options:
  -4, --ipv4          only ping IPv4 addresses
  -6, --ipv6          only ping IPv6 addresses
  -b, --size=BYTES    amount of ping data to send, in bytes (default: 56)
  -B, --backoff=N     set exponential backoff factor to N (default: 1.5)
  -c, --count=N       count mode: send N pings to each target
  -f, --file=FILE     read list of targets from a file (- means stdin)
  -g, --generate      generate target list (only if no -f specified)
                     (give start and end IP in the target list, or a CIDR address)
                     (ex. fping -g 192.168.1.0 192.168.1.255 or fping -g 192.168.1.0/24)
  -H, --ttl=N         set the IP TTL value (Time To Live hops)
  -I, --iface=IFACE   bind to a particular interface
  -l, --loop          loop mode: send pings forever
  -m, --all           use all IPs of provided hostnames (e.g. IPv4 and IPv6), use with -A
  -M, --dontfrag      set the Don't Fragment flag
  -O, --tos=N         set the type of service (tos) flag on the ICMP packets
  -p, --period=MSEC   interval between ping packets to one target (in ms)
                     (in loop and count modes, default: 1000 ms)
  -r, --retry=N       number of retries (default: 3)
  -R, --random         random packet data (to foil link data compression)
  -S, --src=IP        set source address
  -t, --timeout=MSEC  individual target initial timeout (default: 500 ms,
                     except with -l/-c/-C, where it's the -p period up to 2000 ms)

Output options:
  -a, --alive         show targets that are alive
  -A, --addr          show targets by address
  -C, --vcount=N      same as -c, report results in verbose format
  -d, --rdns          show targets by name (force reverse-DNS lookup)
  -D, --timestamp     print timestamp before each output line
  -e, --elapsed       show elapsed time on return packets
  -I, --interval=MSEC interval between sending ping packets (default: 10 ms)
  -n, --name          show targets by name (reverse-DNS lookup for target IPs)
  -N, --netdata       output compatible for netdata (-l -Q are required)
  -o, --outage        show the accumulated outage time (lost packets * packet interval)
  -q, --quiet         quiet (don't show per-target/per-ping results)
  -Q, --quiet=SECS    same as -q, but add interval summary every SECS seconds
  -s, --stats         print final stats
  -u, --unreach       show targets that are unreachable
  -v, --version       show version
  -x, --reachable=N   shows if >=N hosts are reachable or not

(END)_
```

- ```
fping -c 100 192.168.56.3
```

[illegible]

- ```
fping microsoft.com ebay.com citibank.com google.com slashdot.org
yahoo.com
```

```
(kali㉿kali)~  
$ fping microsoft.com ebay.com citibank.com google.com slashdot.org yahoo.com  
[<- 104.94.222.171]microsoft.com is alive  
microsoft.com : duplicate for [0], 64 bytes, 42.9 ms [<- 95.100.104.10]  
microsoft.com : duplicate for [0], 64 bytes, 42.9 ms  
[<- 209.85.203.113]ebay.com is alive  
[<- 104.18.5.215]google.com is alive  
[<- 74.6.143.26]slashdot.org is alive  
[<- 74.6.143.26]citibank.com is alive  
citibank.com : duplicate for [0], 64 bytes, 100 ms  
yahoo.com is alive
```

- All targets are alive

When to use ARP scan

- ```
sudo netdiscover -i eth0 -r 192.168.56.0/24
```

| IP           | At MAC Address    | Count | Len | MAC Vendor / Hostname |
|--------------|-------------------|-------|-----|-----------------------|
| 192.168.56.1 | 66-7b-3a-8b-64    | 1     | 68  | Unknown vendor        |
| 192.168.56.3 | 48-94-27-6d-5d-11 | 1     | 68  | PCB Systemcncvll Gith |

```

NAME active/passive ARP reconnaissance tool
SYNOPSIS
 nmapscan [-i hostname] [-s scan] [-iL ip] [-g ip] [-f file] [-F file]
 [-L list] [-c count] [-n nodes] [-h]

DESCRIPTION
 nmapscan is an active/passive ARP reconnaissance tool. Initially developed to gain information about wireless networks without DHCP servers in scanning scenarios, it can be used on switched networks. Built on top of libnet and libpcap, it can passively detect online hosts or sniff for specific IP addresses and ARP requests.

 Furthermore, it can be used to inspect your network's ARP traffic, or find network addresses using auto scan mode, which will scan for common local networks.

OPTIONS
 -a string
 The network interface to sniff and inject packets. If no interface is specified, first available will be used.

 -F range
 Scan a range instead of auto scan. range will range value including, for example: 192.168.0.0/24, 192.168.2.0/23, 192.168.0.0/12, 192.168.0.0. Note that nmapscan is also a passive scanner, and all detected IPs will be displayed, in addition to those included in the specified range.

 -f file
 Scan ranges contained in the given file. It must contain only one range per line.

 -o enable passive mode. In passive mode, nmapscan does not send anything, but does only sniff.

 -t file
 Scan a list of known MACs and host names.

 -F filter
 Contains pass filter expression (default: "no").

 -t time
 Sleep given time in milliseconds between each ARP request injection. (default: 1)

 -c count
 Number of times to send each ARP request. Useful for networks with packet loss, so it will scan given times for each host. (default: 1)

 -n node
 List IP of the source IP used for scanning. You can change it if the default host (x.x.x.x/24) is already used. (allowed range is 2 to 253, default: 0)

 -i ignore
 Ignore configuration files at how to find autoscanner and fast mode only. This will use default ranges and IPs for autoscanner and fast mode. See below for information about configuration files.

 manual page: nmapscan.1 - files & how to help on a device.

```



TryHackMe

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[Practice](#)

[Compete](#)



[Go Premium](#)

25 



[← Back to all walkthroughs](#)



## Nmap

An in depth look at scanning with Nmap, a powerful network scanning tool.

  50 min  536,630 

[Share your achievement](#)

[Start AttackBox](#)

[Save Room](#)

[20196 Recommend](#)

[Options](#)

Room completed ( 100% )

Task 1  Deploy

Task 2  Introduction

Task 3  Nmap Switches

Task 4  [Scan Types](#) Overview

Task 5  [Scan Types](#) TCP Connect Scans

Task 6  [Scan Types](#) SYN Scans

Task 7  [Scan Types](#) UDP Scans

☒ Complete the Nmap Room on THM

# Nmap Live Host Discovery

The screenshot shows the TryHackMe interface for the 'Nmap Live Host Discovery' room. The header includes the TryHackMe logo, navigation links (Dashboard, Learn, Practice, Compete), a search bar, a 'Go Premium' button, and a user profile with 26 points. The room title 'Nmap Live Host Discovery' is prominently displayed, along with a description: 'Learn how to use Nmap to discover live hosts using ARP scan, ICMP scan, and TCP/UDP ping scan.' Below this, it indicates the room duration is 120 minutes and has been completed by 256,723 users. A green bar at the top of the task list states 'Room completed (100%)'. The task list consists of seven tasks, all marked as completed with green checkmarks: Task 1: Introduction, Task 2: Subnetworks, Task 3: Enumerating Targets, Task 4: Discovering Live Hosts, Task 5: Nmap Host Discovery Using ARP, Task 6: Nmap Host Discovery Using ICMP, and Task 7: Nmap Host Discovery Using TCP and UDP. Action buttons at the top include 'Share your achievement', 'Start AttackBox', 'Save Room', '5047 Recommend', and 'Options'.

☒ Complete the Nmap Live Host Discovery Room on THM

# Learning to use RustScan

The screenshot shows the TryHackMe interface for the 'RustScan' room. The header is identical to the previous screenshot. The room title 'RustScan' is displayed, with the description: 'Learn how to use RustScan.' It indicates a duration of 45 minutes and 14,750 completions. A green bar at the top of the task list states 'Room completed (100%)'. The task list consists of seven tasks, all marked as completed with green checkmarks: Task 1: About RustScan., Task 2: Installing RustScan, Task 3: Accessible, Task 4: Fast, Task 5: Extensible, Task 6: Adaptive, and Task 7: Scanning Time!. Action buttons at the top include 'Share your achievement', 'Start AttackBox', 'Save Room', '624 Recommend', and 'Options'.

☒ Complete the RustScan Room on THM